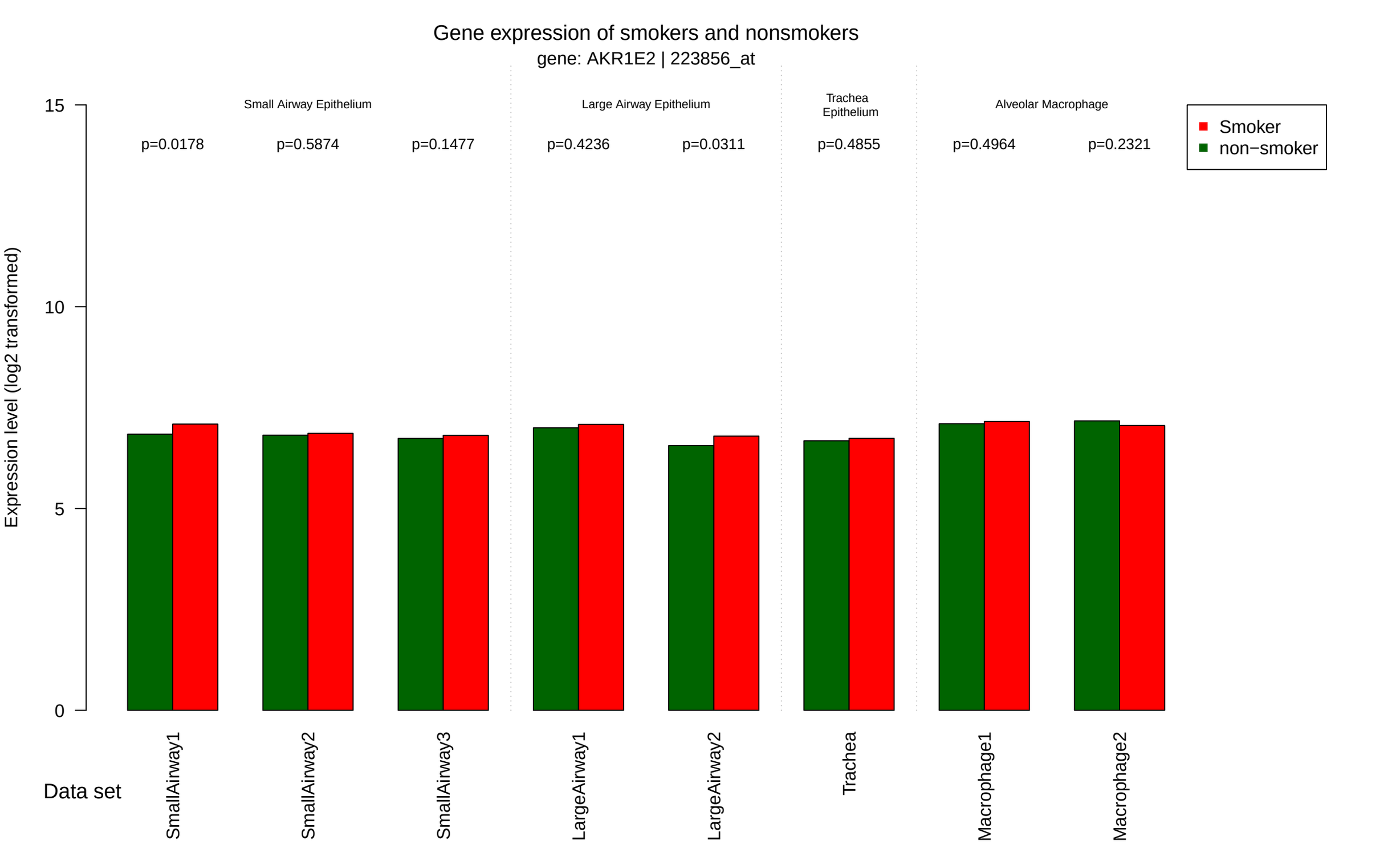
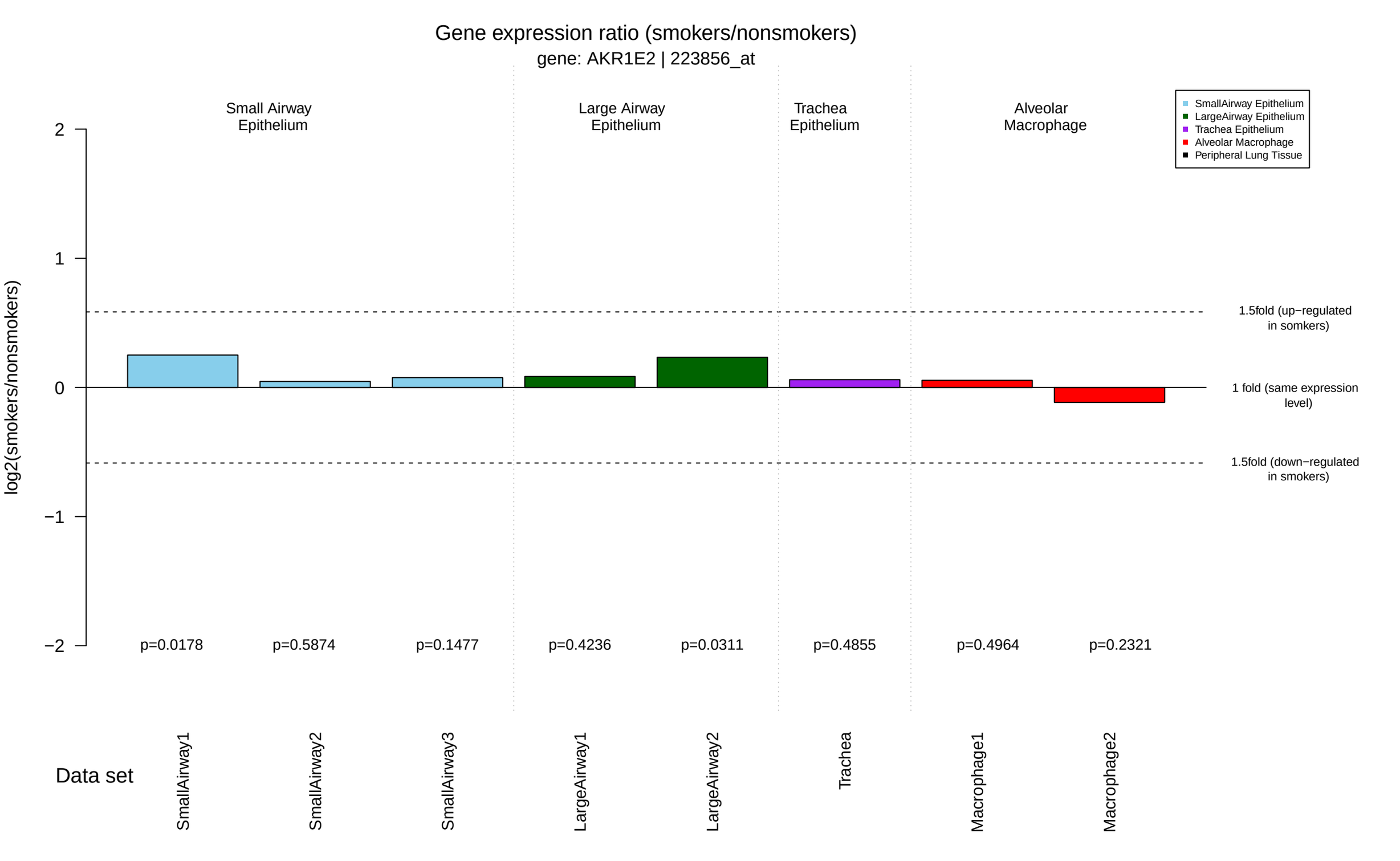
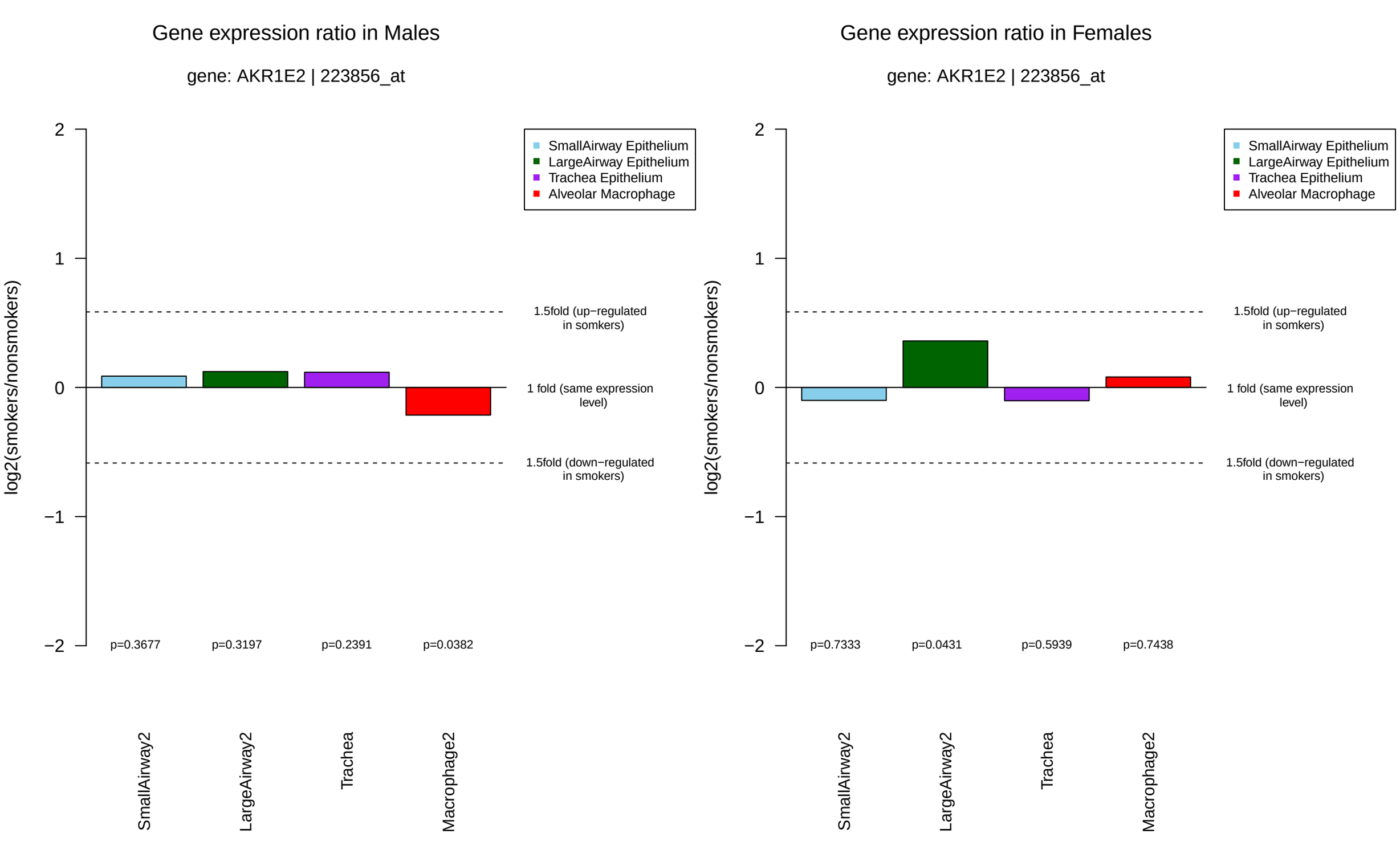
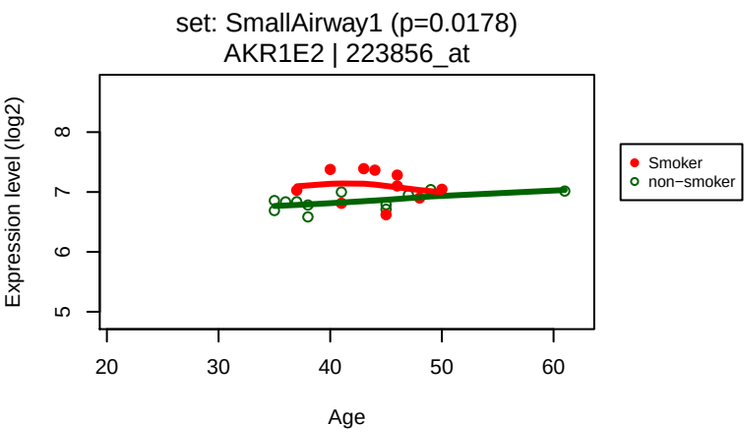
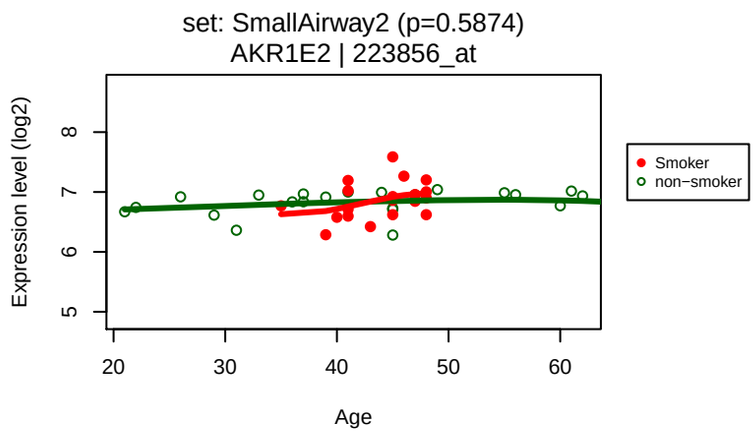
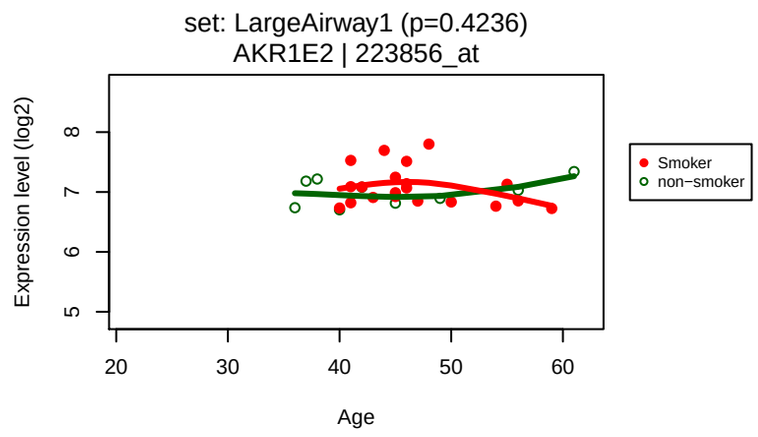

Gene expression of smokers and nonsmokers
gene: AKR1E2 | 223856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0178
p=0.5874
p=0.1477
p=0.4236
p=0.0311
p=0.4855
p=0.4964
p=0.2321
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

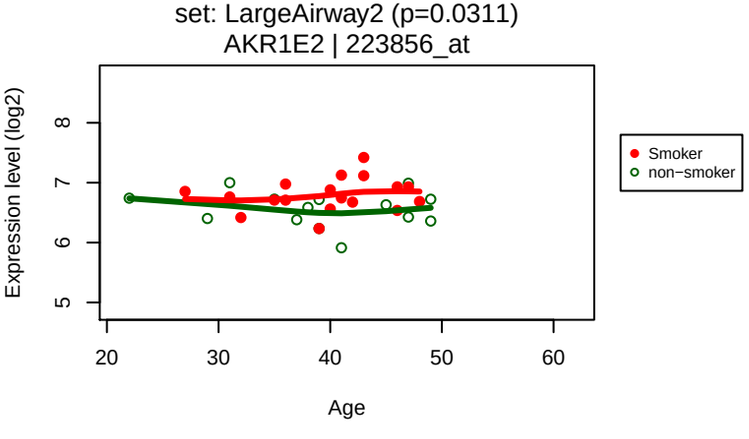

Gene expression ratio (smokers/nonsmokers)
gene: AKR1E2 | 223856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0178
p=0.5874
p=0.1477
p=0.4236
p=0.0311
p=0.4855
p=0.4964
p=0.2321
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

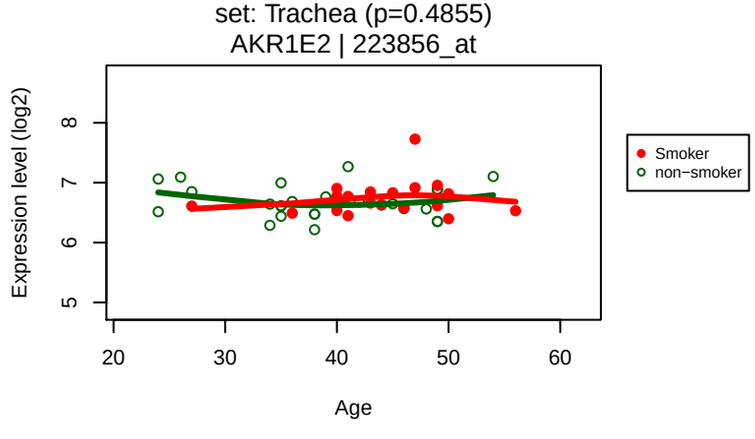

Gene expression ratio in Males
gene: AKR1E2 | 223856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3677
p=0.3197
p=0.2391
p=0.0382
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1E2 | 223856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7333
p=0.0431
p=0.5939
p=0.7438
SmallAirway2
LargeAirway2
Trachea
Macrophage2

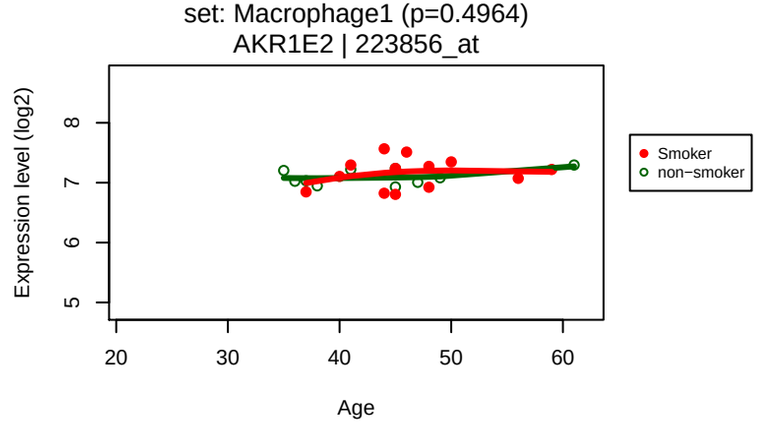

set: SmallAirway1 (p=0.0178)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker

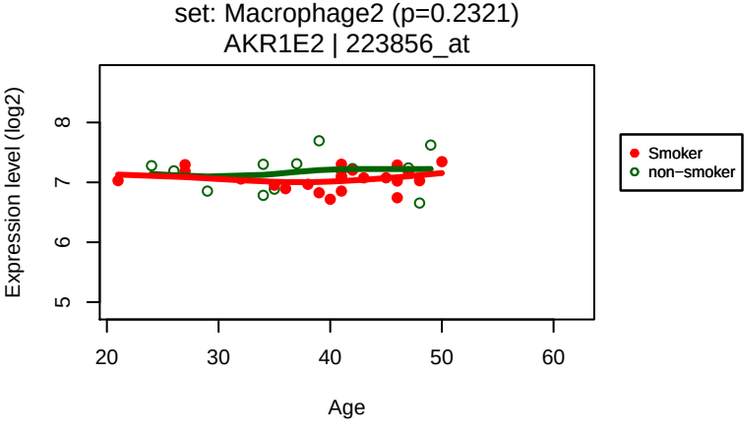

set: SmallAirway2 (p=0.5874)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker

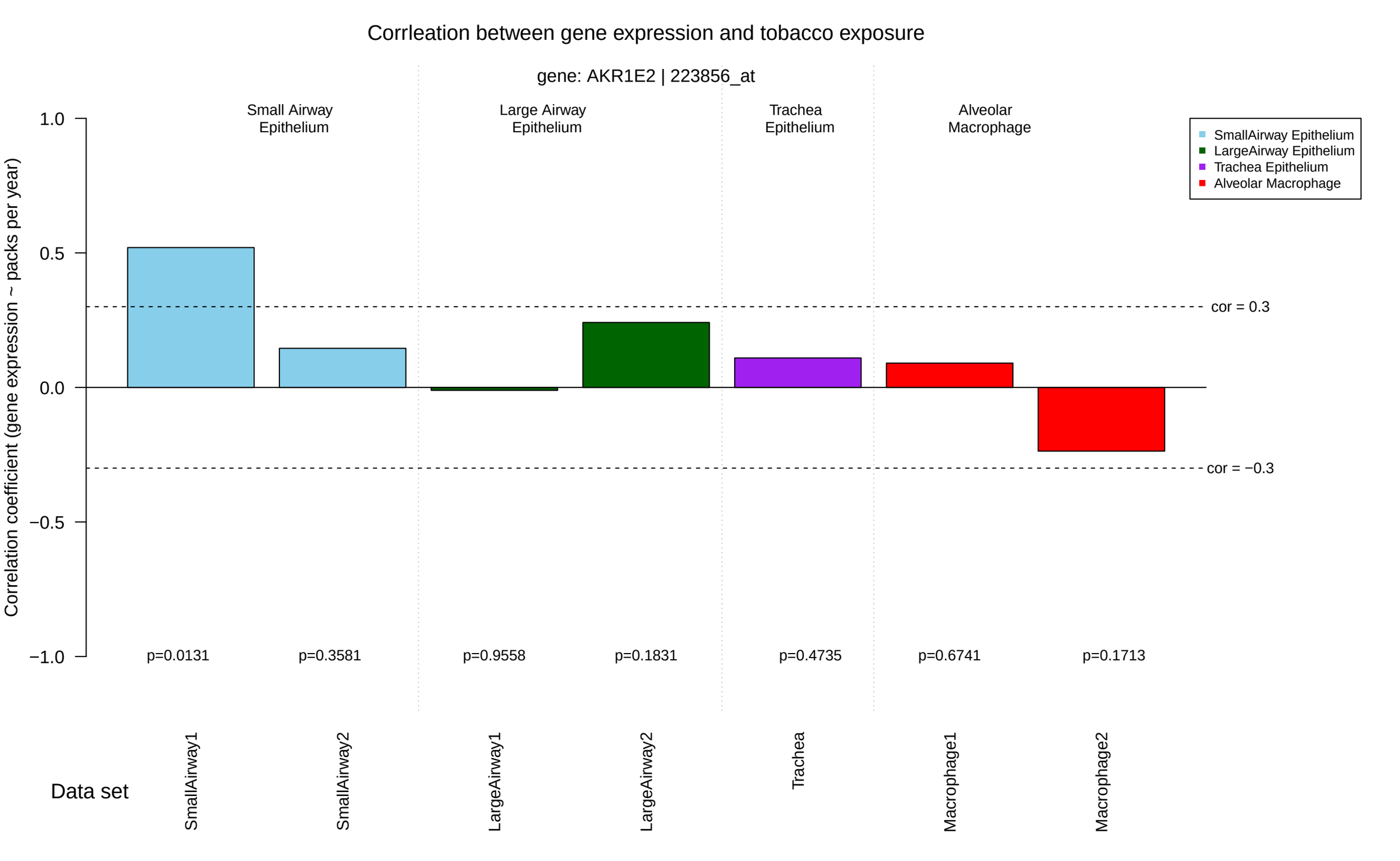

set: LargeAirway1 (p=0.4236)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0311)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4855)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4964)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2321)
AKR1E2 | 223856_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1E2 | 223856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0131
p=0.3581
p=0.9558
p=0.1831
p=0.4735
p=0.6741
p=0.1713
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2